

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1380

Monfort 230 kV
July 26, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1380. The Interconnection Customer proposes a 343.8 MW wind facility interconnecting at the Monfort 230 kV substation in the Indiana area. The request was received on January 10, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 429 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,128,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Illinois Rd – Southport 100kV line	Indego – Columbia Hills 100kV	136.9	127	5.4
Columbia Hills – Portage Uppco 100kV line	Waynedale – Monfort 230kV	132.4	143	11.1
Hart – Southport Substation 100kV	Columbia Hills – Indego 100kV line	129.1	152	5.4
Waynedale – Indego 100kV Line	Monfort – Waynedale 230kV	124.2	242	13.5
Hart – Portage Uppco 100kV	Indego Substation – Columbia Hills 100kV	123.5	177	5.4
Columbia Hills – Indego Substation 100kV	Waynedale – Monfort 230kV Line	111.4	294	13.4
Mckinley – Hebron Rosemeade Substation 100kV L	Monfort Substation – Waynedale 230kV	103.0	134	5.5
Illinois Road – Hebron Rosemeade 100kV	Waynedale – Indego 100kV Line	101.2	163	6.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Illinois Rd – Southport 100kV line	Rebuild line with higher-capacity conductor	\$5,247,000	7% (\$387,000)
Columbia Hills – Portage Uppco 100kV	Rebuild line with higher-capacity conductor	\$2,865,000	12% (\$357,000)
Hart – Southport Substation 100kV	Rebuild line with higher-capacity conductor	\$3,852,000	7% (\$284,000)
Waynedale – Indego 100kV Line	Rebuild line with higher-capacity conductor	\$3,076,000	11% (\$348,000)
Hart – Portage Uppco 100kV	Rebuild line with higher-capacity conductor	\$3,242,000	7% (\$239,000)
Columbia Hills – Indego Substation 100kV	Rebuild line with higher-capacity conductor	\$3,505,000	11% (\$392,000)
Hebron Rosemeade – Mckinley 100kV L	Reconductor line	\$795,000	10% (\$82,000)
Illinois Road – Hebron Rosemeade 100kV	Reconductor line	\$578,000	7% (\$39,000)
	Total allocated to project		\$2,128,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.